

Quantum system quizzing for qudit systems

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Quantum System Quizzing (QSQ) is a recently introduced framework for certification of quantum devices through testing deterministic input–output correlations, rendering it both sample-efficient and user-friendly. Moreover, it operates in a way that is agnostic to state-preparation and measurement (SPAM), assuming only the total dimension of the underlying Hilbert space. Importantly, it comes with soundness guarantees, being able to definitely rule out incorrect implementations of a gate set. So far, sound QSQ-protocols have been developed only in (multi-) qubit architectures exploiting the binary computational logic. In this work, we extend the QSQ framework to certify gate sets implementing the Clifford group for arbitrary odd local dimension. In doing so, we encounter additional, unavoidable discrete symmetries, linked to field automorphisms, emerging from the deterministic nature of the QSQ-protocol. Our results substantially broaden the applicability of QSQ to also cover native qudit quantum computing architectures and give an improved understanding of the mechanisms behind sound certification within the QSQ-framework.

1 Introduction

The continued progress in the development of quantum hardware demands efficient, user-friendly, and rigorous certification methods to evaluate gate performance [1–4]. In this regard, the issue of errors related to state preparation and measurement (SPAM) poses a significant challenge. Several methods have been proposed in the literature to address this problem, such as randomized benchmarking (RB) [5–8] and gate set tomography (GST) [9–12], which are well-established methods and have been developed extensively over the past decades. Recently, some of the authors proposed a new certification method called *quantum system quizzing* (QSQ). It has several desirable properties, notably being sample-efficient, operating in a state prepa-

ration and measurement (SPAM)-agnostic manner, and having a straightforward implementation. Unlike other established quantum certification, validation and verification (QCVV) methods, such as RB and GST, it also directly links experimental observations to meaningful figures of merit, avoiding the need to discuss non-physical gauge transformations. Most importantly, because of this, it is able to provide *sound* certification, i.e., it has the ability to rule out incorrect implementations. At the core, this feature emerges from a separation in the memory requirements between classical and quantum finite state automata to solve certain computational tasks, so-called *promise-problems* [13, 14]. Similar to the maximal violation of certain Bell inequalities achieving what is commonly known as *self-testing* [15, 16], this separation singles out the underlying system by observing special extremal correlations in many interesting cases. QSQ as a certification protocol has been thoroughly analyzed for the simplest case of a single-qubit quarter-rotation [17, 18], with analytical proofs for an inverse linear scaling between the sample complexity and the targeted fidelity. On the theoretical side, it has also been generalized to more general qubit gate sets [19] and to multi-qubit systems. The latter was achieved by reducing the problem to the certification of the individual subsystems first, effectively establishing a way to certify local addressability of subsystems via input-output correlations in a dimension-constrained scenario. In this work, we extend the theory in yet another direction. All previous results made explicit use of the qubit structure of (sub-)systems and several of the core arguments break down when considering higher-dimensional native gate sets. Here, we therefore set out to identify new ways of obtaining similar results to widen the applicability of QSQ. As a byproduct we develop a unified understanding of symmetries that are inherently tied to the nature of the QSQ framework.

This manuscript is organized as follows: We begin with setting up the notation and revisit the notion of quantum models with related definitions. We then proceed to reiterate the QSQ-protocol before moving on to qudit quantum computation, the related gates and introduce the models, for which we will

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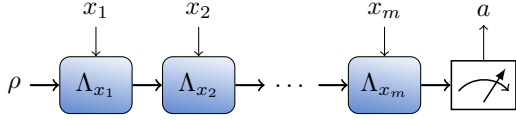


Figure 1: The considered scenario: a quantum system is initialized, undergoes a series of transformations, specified by classical instructions $x_1, x_2, \dots, x_m$, chosen from a finite set Σ , and is measured producing an outcome $a \in A$.

construct QSQ quizzes. The discussion continues with the discrete symmetries which arise as automorphisms of intermediate fields $\mathbb{Q} \subset K \subset \mathbb{C}$ over which the quantum models can be expressed under suitable basis changes. Following this, we present our results for single-qudit systems, before also verifying that the ideas of reducing certification to subsystems as outlined in [19] is applicable also when the formerly identified discrete symmetries of subsystems are non-trivial.

2 Preliminaries & Notation

The sets of density and unitary operators on a Hilbert space $\mathcal{H}$ are denoted by $\mathcal{D}(\mathcal{H})$ and $\mathcal{U}(\mathcal{H})$, respectively. The Pauli matrices are denoted by X, Y and Z , with their corresponding eigenstates given by $|\pm\rangle, |\pm_y\rangle, \{|0\rangle, |1\rangle\}$, respectively. We denote the Hadamard gate by $H = (X + Z)/\sqrt{2}$ and the S -gate by $S = |0\rangle\langle 0| + i|1\rangle\langle 1|$. We use a superscript $U^{(i)}$ to denote the canonical embedding of the unitary U acting on qubit i to the n -qubit space. For pure states $|\psi\rangle$, we often denote the corresponding projector in the mixed-state formalism by ψ . For equivalences between states or unitaries that hold projectively, that is, up to a phase, we use the notation $\overset{\circ}{=}$. The primary goal of this study is to certify the group structure of the gates which are most commonly used in qudit architectures [20–22].

2.1 Qudit Quantum computing

For a d -dimensional system, the Pauli operators generalize to the Weyl-Heisenberg operators. The Weyl-Heisenberg group $\mathcal{W}_d$ is generated by the shift and phase operators $X_d := \sum_{k \in [d]} |k \oplus_d 1\rangle\langle k|$ and $Z_d := \sum_{k \in [d]} \omega^k |k\rangle\langle k|$, respectively, where $\omega = e^{2\pi i/d}$. Here and throughout, $\oplus_d$ denotes addition modulo d . Whenever the dimension is clear from context, we omit the subscript d . The generators satisfy the Weyl commutation relation $X_d Z_d = \omega^{-1} Z_d X_d$, as well as $X_d^d = Z_d^d = \mathbb{1}$. Consequently, every element of $\mathcal{W}_d$ is of the form $\omega^r X_d^p Z_d^q$ for some $p, q, r \in [d]$. Thus, modulo the central phases gen-

erated by $\omega \mathbb{1}$, every Weyl-Heisenberg operator has a unique representative $W(p, q) = X^p Z^q$, for some $p, q \in [d]$.

When d is prime, the projective Weyl-Heisenberg operators can be organized into $(d + 1)$ maximal commuting cyclic subgroups. More precisely, defining $\mathcal{W}'_d := \mathcal{W}_d / \langle \omega \mathbb{1} \rangle$ and denoting the projective class of an operator U by $[U]$, we have

$$\mathcal{W}'_d = \langle [Z_d] \rangle \cup \bigcup_{a \in [d]} \langle [X_d Z_d^a] \rangle, \quad (1)$$

where the subgroups intersect only in the identity element. It is straightforward to verify that the ω^j -eigenstate $|a; j\rangle$ of the cyclic generators $X Z_d^a$ can be expressed in terms of the computational basis (Z_d -eigenbasis) via

$$|a; j\rangle = \frac{1}{\sqrt{d}} \sum_{k \in [d]} \omega^{-jk + ak(k-1)/2} |k\rangle. \quad (2)$$

Accordingly, we denote the eigenbases as $\mathcal{B}_a = \{|a; 0\rangle, \dots, |a; d-1\rangle\}$ and $\mathcal{B}_\infty = \{|k\rangle\}_{k \in [d]}$ for the computational basis. For the X -eigenbasis $\mathcal{B}_0$ we usually denote the states as $|\omega^j\rangle := |0; j\rangle$. These eigenbases have an important geometric feature: each basis $\mathcal{B}_a$ is mutually unbiased with respect to the computational basis $\mathcal{B}_\infty$,

$$|\langle k | a; j \rangle|^2 = \frac{1}{d} \quad \text{for all } a, j, k \in [d]. \quad (3)$$

Moreover, when d is an odd prime, any two distinct bases $\mathcal{B}_a$ and $\mathcal{B}_b$ are mutually unbiased. Consequently, $\{\mathcal{B}_\infty, \mathcal{B}_0, \dots, \mathcal{B}_{d-1}\}$ forms a complete set of $(d + 1)$ mutually unbiased bases [23].

The qudit Clifford group

As in the qubit case, the projective Clifford group is defined as the normalizer of the Weyl-Heisenberg group in the unitary group, modulo global phases,

$$\mathcal{Cl} := \{U \in \mathcal{U}(d) | U \mathcal{W} U^\dagger = \mathcal{W}\} / \mathcal{U}(1). \quad (4)$$

For a single qudit of odd dimension, the Clifford group is generated by the Fourier gate and the quadratic phase gate,

$$F_d := \frac{1}{\sqrt{d}} \sum_{j, k \in [d]} \omega^{jk} |k\rangle\langle j|, \quad (5)$$

$$Q_d := \sum_{j \in [d]} \omega^{j(j-1)/2} |j\rangle\langle j|. \quad (6)$$

We use the symbol Q to emphasize the quadratic dependence of its phases on the computational-basis label.

An important difference from the qubit case is that the Fourier gate satisfies $F_d^2|j\rangle = |-j\rangle$, where the label $-j$ is understood modulo d . Thus, F_d^2 implements the parity transformation. For $d = 2$, this reduces to the familiar identity $H^2 = \mathbb{1}$, since $-j = j$ modulo 2. For odd d , the Weyl-Heisenberg generators can be recovered from F and Q through

$$Z = Q^{d-1}F^2QF^2 \quad \text{and} \quad X = F^3ZF. \quad (7)$$

The Clifford group permutes the eigenstates of the maximal commuting Weyl-Heisenberg subgroups, commonly called stabilizer states, and therefore is not universal for quantum computing. For completeness, we summarize the action of X, Z, F, Q on the states $|a; j\rangle$ in Lemma 11 for prime d .

In odd prime dimension, adjoining any non-Clifford unitary to the Clifford group produces an approximately universal gate set. For $d \geq 5$, examples of such non-Clifford gates are provided by computational-basis permutations

$$P_\pi = \sum_j |\pi(j)\rangle\langle j|, \quad \pi \in \mathfrak{S}_d, \quad (8)$$

where π is *not* an affine transformation of the form $\pi(j) = aj + b \pmod d$ for some $a \in \mathbb{Z}_d^\times, b \in \mathbb{Z}_d$.

2.2 Quantum Models

Throughout, we consider the computational scenario illustrated in Fig. 1: A quantum system is prepared in an initial state ρ , undergoes a sequence of transformations $\Lambda_{\mathbf{x}}$ specified by an input string $\mathbf{x} = x_1 \cdots x_l \in \Sigma^*$, and is finally measured M_a , producing an outcome $a \in A$. Here, Σ^* denotes the set of all finite strings over the instruction alphabet Σ , and A denotes the set of possible measurement outcomes. For $\mathbf{x} \in \Sigma^*$, we write $\mathbf{x}^k$ for its k -fold repetition, with the convention that $\mathbf{x}^0 = \epsilon$ is the *empty string*.

We assume that every occurrence of an instruction $x \in \Sigma$ is represented by the same quantum channel Λ_x , independently of its position in the input string and of the other instructions appearing in the experiment. We refer to this assumption as *context independence*. Under this assumption, the experiment is described by a *quantum model*.

A quantum model [24] is a tuple $\mathcal{M} = (\rho, \{\Lambda_x\}_{x \in \Sigma}, \{M_a\}_{a \in A})$ defined on a finite-dimensional Hilbert space $\mathcal{H}$. Here, $\rho \in \mathcal{D}(\mathcal{H})$ is the initial state, each Λ_x is a quantum channel on $\mathcal{H}$, and $M_{a \in A}$ is a POVM on $\mathcal{H}$. If $\rho = |\phi\rangle\langle\phi|$ is pure, every channel is unitary, with $\Lambda_x(\cdot) = U_x(\cdot)U_x^\dagger$, and the measurement is the projective measurement associated with an orthonormal basis $\mathcal{B} = |b_a\rangle_{a \in A}$,

we write $\mathcal{M} = (|\phi\rangle, \{\Lambda_x\}_{x \in \Sigma}, \mathcal{B})$ and call $\mathcal{M}$ a *unitary model*.

Let Σ' be a set of additional instruction labels disjoint from Σ , with associated channels $\{\Lambda_x\}_{x \in \Sigma'}$. We denote the model obtained by augmenting the gate set of $\mathcal{M}$ by $\mathcal{M} + \{\Lambda_x\}_{x \in \Sigma'} = (\rho, \{\Lambda_x\}_{x \in \Sigma \cup \Sigma'}, \{M_a\}_{a \in A})$.

Whenever a map f has a specified action on states, channels, and measurement effects, we use the component-wise shorthand notation $f(\mathcal{M}) := (f(\rho), \{f(\Lambda_x)\}, \{f(M_a)\})$. This notation does not, by itself, imply that $f(\mathcal{M})$ is a valid quantum model; its physicality must be verified separately. When f acts entrywise, its action is understood with respect to a fixed basis and, for channels, a fixed superoperator representation. Two quantum models $\mathcal{M} = (\rho, \{\Lambda_x\}_{x \in \Sigma}, \{M_a\}_{a \in A})$ and $\mathcal{N} = (\tilde{\rho}, \{\tilde{\Lambda}_x\}_{x \in \Sigma}, \{\tilde{M}_a\}_{a \in A})$, both defined over $\mathcal{H}$, are called *equivalent* (denoted as $\mathcal{M} \sim_u \mathcal{N}$), if there exists a unitary U such that $U\tilde{\rho}U^\dagger = \rho$, $U\tilde{M}_aU^\dagger = M_a$ for all $a \in A$ and $U\tilde{\Lambda}_x(U^\dagger(\cdot)U)U^\dagger = \Lambda_x(\cdot)$ for all $x \in \Sigma$.

Unitary equivalence amounts to a global change of orthonormal basis and therefore cannot be distinguished operationally. More generally, Wigner's theorem identifies unitary and antiunitary transformations as the fundamental symmetries of probabilities in quantum mechanics [25]. After quotienting by unitary transformations, the antiunitary freedom may be represented by entrywise complex conjugation in a fixed basis. Unlike Refs. [13, 17–19], we do not include complex conjugation in the equivalence relation ($\sim_u$). For the models considered here, the support-only nature of the QSQ-protocol can give rise to a broader discrete family of ambiguities associated with field automorphisms, with complex conjugation as a distinguished instance. These automorphisms are not general symmetries of quantum mechanics and are considered only when their action produces a valid quantum model.

To formulate quantitative certification statements, we introduce a measure of discrepancy between quantum models. For states, we use the state infidelity $1 - F(\rho, \tilde{\rho})$ [26], while for channels we use the average gate infidelity $1 - F_{\text{avg}}(\Lambda, \tilde{\Lambda})$ [27]. To compare two POVMs, we maximize the total variation distance between the outcome distributions they induce over all possible input states [28]. For two models with the same instruction and outcome labels, we then define the model infidelity

$$\text{inFid}^*(\mathcal{M}, \mathcal{N}) := \max\{1 - F(\tilde{\rho}, \rho), 1 - \min_x F_{\text{avg}}(\tilde{\Lambda}_x, \Lambda_x), d_{TV}(\{M_a\}, \{\tilde{M}_a\})\}. \quad (9)$$

To account for unitary equivalence, we minimize this quantity over all unitary gauges of the second model,

$$\text{inFid}(\mathcal{M}, \mathcal{N}) := \min_{\mathcal{N}' \sim_u \mathcal{N}} \text{inFid}^*(\mathcal{M}, \mathcal{N}'). \quad (10)$$

Although we use the notation inFid , this quantity is a gauge-optimized model infidelity and is not asserted to satisfy all metric axioms.

Finally, due to the support-based nature of the QSQ-protocol, we introduce also the *output map* of a model $\mathcal{M}$, which is defined by assigning to each possible input sequence $\mathbf{x} \in \Sigma^*$ the set of measurement outcomes that appear with non-zero probability.

$$\mathbf{a}_{\mathcal{M}}(\mathbf{x}) := \{a \in \mathcal{A} \mid \text{Tr}[M_a \Lambda_{x_l} \circ \dots \circ \Lambda_{x_1}(\rho)] > 0\}. \quad (11)$$

With these definitions in place, we can now formally state the QSQ-protocol.

2.3 Quantum system quizzing

The QSQ protocol is based on a very simple idea of testing the output behavior of a device under specific input sequences (questions) in the spirit of *direct certification* [1]. It operates by testing the plausibility of randomly chosen input instructions against some target model $\mathcal{M}$, as specified via its output map $\mathbf{a}_{\mathcal{M}}$.

Algorithm 1 Quantum system quizzing (QSQ)

Input: N – number of repetitions, $\mathcal{X} \subset \Sigma^*$ – set of gate sequences, $\mathbf{a}_{\mathcal{M}}$ – the output map for a target model $\mathcal{M}$.
for $i \in \{1, \dots, N\}$ **do**
 sample $\mathbf{x}$ from $\mathcal{X}$;
 run the quantum circuit corresponding to $\mathbf{x}$,
 record the outcome a ;
 if $a \notin \mathbf{a}_{\mathcal{M}}(\mathbf{x})$ **then**
 output "reject" and end the protocol;
 output "accept".

Each run of the QSQ protocol starts with an input $\mathbf{x}$ selected at random from the predetermined set of sequences, or *quizzes*, $\mathcal{X}$, after which a quantum system is initialized, the sequence of gates corresponding to $\mathbf{x}$ is performed, the system is measured with the outcome a recorded. If the response is not among the measurement outcomes $\mathbf{a}_{\mathcal{M}}(\mathbf{x})$ expected from the target quantum model $\mathcal{M}$, the protocol aborts by rejecting the implemented quantum model.

Clearly, the QSQ-protocol leads to a *complete* test: if the implemented model $\mathcal{N}$ is equivalent to the

target model $\mathcal{M}$, the protocol outputs ‘accept’ for any N . Conversely, suppose that the QSQ-protocol accepts $\mathcal{N}$ with probability one for every N . Then no outcome outside the support predicted by $\mathcal{M}$ can occur with positive probability for any tested input sequence. Consequently, $\mathbf{a}_{\mathcal{N}}(\mathbf{x}) \subseteq \mathbf{a}_{\mathcal{M}}(\mathbf{x})$ for every $\mathbf{x} \in \mathcal{X}$.

The central question is whether this support condition is sufficiently restrictive to identify the target model. For certain target models, some of us have shown that one can construct a finite set of input sequences $\mathcal{X}$ such that, together with an appropriate dimension constraint, the inclusions $\mathbf{a}_{\mathcal{N}}(\mathbf{x}) \subseteq \mathbf{a}_{\mathcal{M}}(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{X}$ imply that $\mathcal{N}$ is unitarily equivalent either to $\mathcal{M}$ or to its complex conjugate [17, 19]. For such models, the protocol is asymptotically sound: every fixed implementation outside this equivalence class produces a forbidden outcome with positive probability for at least one input sequence in $\mathcal{X}$. The probability that the protocol detects this discrepancy therefore approaches one as $N \rightarrow \infty$. A quantitative finite-sample soundness guarantee additionally requires a lower bound on the probability of observing such a forbidden outcome, which we call certification protocol *robustness results*.

The paradigmatic example for QSQ-certifiability is the certification of the quarter-phase rotation on a single qubit.

Example 1 (See Thm. 11 in [19]). *The QSQ-protocol for the model $\mathcal{S} = (|+\rangle, \{\mathcal{S}\}, \{|+\rangle, |-\rangle\})$ with the input sequences $\mathcal{X} = \{\epsilon, s^2, s^4\}$ is sound in the limit of infinitely many rounds.*

In the preceding example, the target model is certified up to unitary equivalence. For more general qubit models, including the single-qubit Clifford model, complex conjugation constitutes an additional unavoidable ambiguity [17].

The primary objective of this work is to establish analogous QSQ certification results for models of the form

$$\begin{aligned} \mathcal{F}_d &:= (|0\rangle, \{X_d, F_d, Z_d\}, \mathcal{B}_\infty) \\ \mathcal{Cl}_d &:= \mathcal{F}_d + \{Q_d\}. \end{aligned} \quad (12)$$

In dimensions d greater than three, we will encounter additional symmetries beyond complex conjugation in the support-based certification of these models. We characterize these ambiguities before turning to the corresponding QSQ certification results.

3 Galois symmetries

In this section, we identify additional ambiguities/symmetries that must be accounted for in QSQ certification. These ambiguities extend beyond unitary equivalence and complex conjugation and arise from the support-based nature of the protocol.

The underlying mechanism is a mismatch between the field (in the algebraic sense) over which a quantum model is defined and the limited information about its probabilities retained by the QSQ-protocol. Here, so-called *field-automorphisms*, i.e. mappings from a field into itself, which respect the field structure, play a central role.

The complex conjugation $\mathcal{I}_{\mathbb{C}}$ provides a familiar example. It is a field automorphism of $\mathbb{C}$ that fixes $\mathbb{R}$ pointwise. Because experimentally observed probabilities are real, simultaneously conjugating all components of a quantum model leaves its probabilities unchanged and therefore constitutes an unavoidable certification ambiguity.

Within the QSQ-framework, this effect becomes more pronounced for two reasons. First, the QSQ-protocol is only based on not observing zero-probability outcomes, therefore *all* automorphisms potentially lead to additional symmetries. Secondly, and equally important, models typically investigated in the context of QSQ admit a description over fields which are smaller than $\mathbb{C}$, such as the cyclotomic fields for $\mathcal{F}_d$, leading to much richer automorphism groups.

Before continuing our discussion, we briefly review the relevant concepts from the theory of field extensions and Galois theory. For definitions, proofs, and further background, we refer to Chapters 5 and 6 of Ref. [29].

3.1 Field extensions & Automorphisms

A field extension is an inclusion of fields $K \subseteq L$, commonly denoted by L/K . The field L then naturally carries the structure of a K -vector space, its dimension is denoted by $[L : K]$ and is called the *degree* of the extension.

In this work, we are primarily concerned with field extensions of $\mathbb{Q}$. These are typically obtained by adjoining an element $\alpha \in \mathbb{C} \setminus \mathbb{Q}$. The notation $\mathbb{Q}(\alpha)$ denotes the smallest subfield of $\mathbb{C}$ containing both $\mathbb{Q}$ and α . If α is a root of a nonzero polynomial $p(t) \in \mathbb{Q}[t]$, then α and the extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ are called *algebraic*; otherwise, α is called *transcendental*. If α is algebraic, then $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is finite and equals the degree of the minimal polynomial of α over $\mathbb{Q}$. In particular, if $p(t)$ is irreducible and has α as a root, then $p(t)$ is the minimal polynomial of α up to

normalization, and $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is equal to the degree of $p(t)$.

As an example, the complex numbers $\mathbb{C}$ are the extension over $\mathbb{R}$ obtained by adjoining i as a root of the irreducible polynomial $t^2 + 1$. Of particular interest for this work are the so-called *cyclotomic fields* $\mathbb{Q}_d := \mathbb{Q}(e^{2\pi i/d})$, which are obtained by adjoining a primitive root of unity of the corresponding order to the rationals.

Given a field extension L/K , we denote by $\text{Aut}_K(L)$ the group of field automorphisms of L that fix every element of K . Since every field automorphism in characteristic zero fixes the rational numbers, $\text{Aut}_{\mathbb{Q}}(K) = \text{Aut}(K)$ whenever $\mathbb{Q} \subseteq K$. For a finite extension L/K , the number of such automorphisms satisfies $|\text{Aut}_K(L)| \leq [L : K]$. If equality holds, the extension is called *Galois*, and $\text{Aut}_K(L)$ is called its *Galois group*, denoted by $\text{Gal}(L/K)$. When the base field is $K = \mathbb{Q}$ we simply write $\text{Gal}(L)$. Informally, a Galois extension has the largest possible number of field symmetries; see [29] for a detailed discussion.

It is well known that the cyclotomic fields $\mathbb{Q}_d$ are such Galois extensions with Galois groups isomorphic to the multiplicative groups $\mathbb{Z}_d^\times$. The action on the roots of unity is simply via exponentiation, so $\tau \in \mathbb{Z}_d^\times \cong \text{Gal}(\mathbb{Q}_d)$ acts by $\tau(\zeta_d) = \zeta_d^\tau$.

3.2 Field Automorphisms in QSQ

Let $\mathcal{M} = (|\psi_0\rangle, \{U_x\}_{x \in \Sigma}, \{|a\rangle\}_{a \in \mathcal{A}})$ be a unitary target model. Fix a basis of $\mathcal{H}$ and express the model in the open-system representation: $|\psi_0\rangle \mapsto |\psi_0\rangle\rangle$, $|a\rangle \mapsto \langle\langle a|$, and $U_x \mapsto U_x \otimes \bar{U}_x$. We define $K_{\mathcal{M}}$ as the smallest intermediate subfield of $\mathbb{C}$ containing $\mathbb{Q}$ and all entries of these vectors and superoperators.

For an input string $\mathbf{x} = x_1 \cdots x_l$, let $U_{\mathbf{x}} := U_{x_l} \cdots U_{x_1}$. The probability of obtaining outcome a is then $\langle\langle a|(U_{\mathbf{x}} \otimes \bar{U}_{\mathbf{x}})|\psi_0\rangle\rangle$. Now observe the following: Every automorphism $\tau \in \text{Aut}_{\mathbb{Q}}(K_{\mathcal{M}})$ acts entrywise on the open-system representation. Since a field automorphism preserves sums, products, and zero, we obtain

$$\begin{aligned} \langle\langle a|U_{\mathbf{x}} \otimes \bar{U}_{\mathbf{x}}|\psi_0\rangle\rangle &= 0 \\ \Leftrightarrow \langle\langle \tau(a)|\tau(U_{\mathbf{x}} \otimes \bar{U}_{\mathbf{x}})|\tau(\psi_0)\rangle\rangle &= \tau(0) = 0. \end{aligned} \quad (13)$$

Thus, whenever the entrywise-transformed components define a valid quantum model, denoted by $\tau(\mathcal{M})$, the two models have the same output map: $\mathbf{a}_{\tau(\mathcal{M})} = \mathbf{a}_{\mathcal{M}}$. Note that a priori any *embedding* $K \hookrightarrow \mathbb{C}$ leads to additional symmetries. However, it is a standard result in algebra that a finite algebraic extension $K/\mathbb{Q}$ admits exactly $[K : \mathbb{Q}]$ such

embeddings, which arise as field automorphisms of a suitable algebraic extension of K . In particular, for a Galois-extension all its embeddings into $\mathbb{C}$ are exactly given by its automorphisms.

To ensure compatibility with complex conjugation, we restrict attention to coefficient fields $K_{\mathcal{M}}/\mathbb{Q}$ that are finite abelian Galois extensions. Complex conjugation is then an element of $\text{Gal}(K_{\mathcal{M}}/\mathbb{Q})$ and, because this group is abelian, commutes with every other field automorphism. As discussed after Proposition 3, this restriction includes the models of primary interest here. For general mixed-state or noisy-channel models, additional conditions would be required because field automorphisms need not preserve positivity or complete positivity. This observation motivates the following notion of QSQ certification, which generalizes the definitions introduced in Refs. [17, 19].

Definition 2. *Let $\mathcal{M}$ be a unitary quantum model over a finite-dimensional Hilbert space $\mathcal{H}$, represented over a finite abelian Galois extension $K/\mathbb{Q}$. Let $G \subset \text{Gal}(K)$, not necessarily a subgroup. We say that a set of input strings $\mathcal{X}$ QSQ-certifies $\mathcal{M}$ up to the action of G (up to G -symmetries), if every quantum model $\mathcal{N}$ over $\mathcal{H}$ with the same instruction and outcome labels as $\mathcal{M}$, satisfying $\mathbf{a}_{\mathcal{N}}(\mathbf{x}) \subseteq \mathbf{a}_{\mathcal{M}}(\mathbf{x})$, for every $\mathbf{x} \in \mathcal{X}$ is unitarily equivalent to $\tau(\mathcal{M})$ for some $\tau \in G$.*

However, the coefficient field $K_{\mathcal{M}}$ generally depends on the basis used to represent the model. Consider, for example, the unitarily equivalent qubit models $\mathcal{M} = (|+\rangle, \{R_{\pi/8}^z\}, \{|+\rangle, |-\rangle\})$ and $\mathcal{N} = (|+\rangle, \{R_{\pi/8}^y\}, \{|+\rangle, |-\rangle\})$ with $R_{\theta}^z = \exp(i\theta Z)$ and $R_{\theta}^y = \exp(i\theta Y)$. Their unitary equivalence is implemented by conjugation with a suitable square root of X , which maps Z to Y while preserving the preparation and measurement projectors. In the open-system representation, their coefficient fields are $K_{\mathcal{M}} = \mathbb{Q}_8$ and $K_{\mathcal{N}} = \mathbb{Q}(\sqrt{2})$. Clearly, these are distinct Galois extensions of $\mathbb{Q}$: the first is not contained in $\mathbb{R}$, whereas the second is totally real. Moreover, they are of different degree, and hence their Galois groups likewise differ, given by $\text{Gal}(K_{\mathcal{M}}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ and $\text{Gal}(K_{\mathcal{N}}) = \mathbb{Z}_2$, respectively. On the other hand, the additional freedom granted in Definition 2 should not rely on the coefficient fields themselves being basis-dependent, as in the above example. Instead, we have to consider the orbit of the target model under the action of the admissible field automorphisms *up to unitary equivalence*. The next result shows that, under the stated Galois assumptions, this combined ambiguity class

is independent of the chosen representation of the target model.

Proposition 3. *Let $\mathcal{M} \sim_u \mathcal{N}$ be unitary quantum models represented over finite abelian Galois extensions $K_{\mathcal{M}}/\mathbb{Q}$ and $K_{\mathcal{N}}/\mathbb{Q}$, respectively. Let $\mathcal{O}_{\mathcal{M}} = \{\tau(\mathcal{M}) \mid \tau \in \text{Gal}(K_{\mathcal{M}})\}$ and $\mathcal{O}_{\mathcal{N}} = \{\tau(\mathcal{N}) \mid \tau \in \text{Gal}(K_{\mathcal{N}})\}$. Then $\mathcal{O}_{\mathcal{M}}$ and $\mathcal{O}_{\mathcal{N}}$ contain the same number of unitary-equivalence classes, and we define the Galois-index of $\mathcal{M}$ as $\text{ind}(\mathcal{M}) := |\mathcal{O}_{\mathcal{M}}/\sim_u|$.*

See Section C for a proof. The significance of this result lies in the implication that in any chosen representation over an abelian Galois extension K , we do not have to consider other equivalences beyond unitary conjugation and $\text{Aut}(K)$, which might correspond to field automorphisms in a different representation over another field L . In particular, any G as in Definition 2 must satisfy $|G| \geq \text{ind}(\mathcal{M})$, and conversely, if $\mathcal{M}$ is QSQ-certifiable, then in Definition 2 we can always find a subset $G \subset \text{Gal}(K)$ with $|G| = \text{ind}(\mathcal{M})$.

Lastly, let us discuss how restrictive it is to require abelian Galois coefficient fields. Every complex projective representation of a finite group is projectively equivalent to one that can be realized over a cyclotomic field [30]. If the initial-state and measurement operators are also defined over cyclotomic fields, then, by passing to a common cyclotomic extension, the entire model can be represented over a single cyclotomic field. This condition is not automatic for arbitrary preparations and measurements. For universal gate sets, which generate infinite groups, the finite-group argument no longer applies. Whether such a model admits a cyclotomic realization depends on the arithmetic properties of the additional gates, including their spectra and the alignment of their eigenbases with the remaining model. Nevertheless, all models considered in this work and in the previous works [13, 17–19] admit natural realizations over cyclotomic fields. Since cyclotomic fields are finite abelian Galois extensions of $\mathbb{Q}$, the assumption made above covers all these models.

3.3 Galois-symmetries for concrete instances

The unitary invariance of the Galois index greatly simplifies to determine it explicitly. Let us first reinterpret the results of Refs. [17, 19] in this framework. The results of Ref. [19] show that the model $\mathcal{S} = (|+\rangle, S, \{|+\rangle, |-\rangle\})$ can be certified up to unitary equivalence alone, without an additional complex-conjugation ambiguity. This can be understood using Proposition 3. Under a suitable unitary basis change, $\mathcal{S}$ can be mapped to an equivalent model

whose open-system representation has only rational entries. Consequently, the model can be represented over $\mathbb{Q}$, and hence $\text{ind}(\mathcal{S}) = 1$.

For the ordinary qubit Clifford model, which contains the S-gate, the coefficient field is $\mathbb{Q}(i) = \mathbb{Q}_4$. Its only nontrivial automorphism is complex conjugation. Since $\mathcal{Cl}_2$ and $\overline{\mathcal{Cl}}_2$ are not unitarily equivalent, the two Galois images represent distinct unitary-equivalence classes, and therefore $\text{ind}(\mathcal{Cl}_2) = 2$.

After adjoining the gate $T = R_{\pi/8}^z$, the coefficient field becomes the eighth cyclotomic field $\mathbb{Q}_8 = \mathbb{Q}(e^{i\pi/4})$. Its Galois group is $\text{Gal}(\mathbb{Q}_8/\mathbb{Q}) \cong \mathbb{Z}_8^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. The automorphism $\zeta_8 \mapsto \zeta_8^5$ fixes the Clifford submodel and maps T to TZ , up to a global phase; these two channels have different spectra and are therefore not unitarily equivalent. The remaining two automorphisms restrict to complex conjugation on the Clifford submodel. Thus, the four Galois images belong to four distinct unitary-equivalence classes, giving $\text{ind}(\mathcal{Cl}_2 + \{T\}) = 4$. This reproduces the ambiguity found in [17]: up to complex conjugation, QSQ certifies either $\mathcal{Cl}_2 + \{T\}$ or $\mathcal{Cl}_2 + \{TZ\}$. Finally, let us consider the models in Eq. (12). Clearly, $\mathcal{F}_d$ is defined over $\mathbb{Q}_d$. Now observe that no unitary U can satisfy $UZU^\dagger \stackrel{\sim}{=} Z^\kappa$ for any $\kappa \neq 1$ while also keeping the computational basis fixed. Therefore we find that $\tau(\mathcal{F}_d) \approx_u \mathcal{F}_d$ for all $1 \neq \tau \in \text{Aut}(\mathbb{Q}_d)$, and in particular $\text{ind}(\mathcal{F}_d) = \varphi(d)$, where $\varphi(d) = |\mathbb{Z}_d^\times|$ is Euler's totient-function. The argument carries over to $\mathcal{Cl}_d$ and models of the form $\mathcal{F}_d + \{P_\pi\}, \mathcal{Cl} + \{P_\pi\}$ to show that the Galois index is $\varphi(d)$ in all cases.

4 Results

The initial Example 1 of certification of the S can be considered as a coincidence for two-dimensional systems. It makes extensive use of pigeonhole-principle-based arguments to establish the coherence properties and identify the spectrum of the implemented unitary, both of which are necessary steps to establish QSQ-soundness. These arguments, however, fail already in $d = 3$. In order to establish certifiability of $\mathcal{F}_d$ and $\mathcal{Cl}_d$ in higher dimensions, the most important step is therefore to develop suitable arguments more systematically.

For this, we exploit the special structure observed between the computational basis $\{|0\rangle, \dots, |d-1\rangle\}$ and the Fourier basis $\{|\omega^0\rangle, \dots, |\omega^{d-1}\rangle\}$ with the respective ‘clock’-operators X_d, Z_d , and their interplay through the Hadamard gate F_d , as indicated in Fig. 2. Here, $\mathcal{F}_d$ serves as the ‘atomic’ model for which we derive certifiability first, and then build

the subsequent schemes for $\mathcal{Cl}$ and universal models on top, comparable to the role of $\mathcal{S}$ in qubits. The key ingredient for establishing QSQ-certifiability of $\mathcal{F}$ is the following lemma on ‘clock’-channels.

Lemma 4. *Let Λ be a channel, which acts as a cyclic shift on an ONB $\mathcal{B} = \{|\psi_0\rangle, \dots, |\psi_{d-1}\rangle\}$, i.e., $\Lambda(\psi_i) = \psi_{i \oplus_d 1}$ for all i , and fixes another pure state ϕ , $\Lambda(\phi) = \phi$. Then we can infer the following:*

1. *The channel Λ is implemented by a unitary U_Λ .*
2. *The eigenbasis of U_Λ and $\mathcal{B}$ are mutually unbiased.*
3. *The eigenvalues of U_Λ are given by the distinct d -th order roots of unity.*

Proof. The first observation is that $\Lambda^\dagger$ acts as the reverse cyclic shift (see Lemma 12) on $\mathcal{B}$, and therefore we have

$$\text{Tr}[\psi_{k-1}\phi] = \text{Tr}[\Lambda^\dagger(\psi_k)\phi] = \text{Tr}[\psi_k\Lambda(\phi)] = \text{Tr}[\psi_k\phi]$$

for any $k \in [d]$. Inductively, we conclude $\text{Tr}[\psi_k\phi] = \frac{1}{d}$ for all k . Since ϕ is assumed to be pure, Λ must be implemented by some unitary U_Λ according to Lemma 13. Besides, the same argument above can be applied to prove that $\mathcal{B}$ and the eigenbasis of U_Λ are mutually unbiased. In order to determine the eigenvalues of U_Λ , we follow an argument of [31]. Because of $\Lambda(\psi_k) = \psi_{k \oplus_d 1}$, $\Lambda(\phi) = \phi$, we can conclude that $\Lambda^d = \mathbb{1}$, and by adjusting the global phase of U_Λ we find that $U_\Lambda^d - \mathbb{1} = 0$. On the other hand, since the $U_\Lambda^k|\psi_0\rangle = |\psi_k\rangle$, $k = 0, \dots, d-1$ are linearly independent, there is no polynomial q of smaller degree such that $q(U_\Lambda) = 0$. By the Cayley-Hamilton theorem [32] and a dimension argument, we conclude that $p(z) = z^d - 1$ is the *characteristic polynomial* of U_Λ . Therefore, the distinct order- d roots of unity are exactly the eigenvalues of U_Λ , which finishes the proof. $\square$

Importantly, the conditions of this lemma can be checked solely from the measurement readouts, without any initial trust in the available operations, other than the dimension. Lemma 4 forms the cornerstone of the following certification result, where we explicitly state the needed input sequences. To give an intuition for how these sequences are used, we depict the transitions inside $\mathcal{F}$ for $d = 3$ in Fig. 2.

Theorem 5 (Certification of $\mathcal{F}_d$). *The input instructions*

$$\mathcal{X}_d := \bigcup_{k=0}^{d-1} \{x^k, x^k z, fz^k f, fz^k x f\} \quad (14)$$

certify $\mathcal{F}_d$ up to $\mathbb{Z}_d^\times \cong \text{Aut}(\mathbb{Q}_d)$ -symmetries.

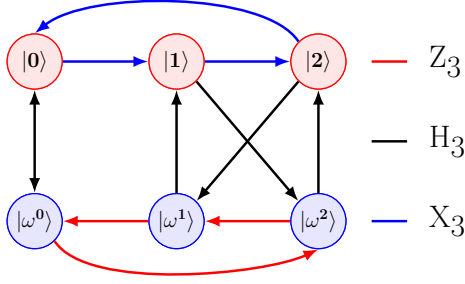


Figure 2: The Fourier-model $\mathcal{F}_3$ on a qutrit. The coloring of the vertices indicate the eigenstates of X, Z .

Proof sketch. We outline the general idea of the proof, and refer to Appendix C for all the details. In the first step, we deduce that there exist ONBs $\mathcal{B}_x, \mathcal{B}_z$ on which the implemented channels Λ_x, Λ_z of the compatible model act as

$$\begin{aligned} \Lambda_x(\psi_k) &= \psi_{k \oplus d1}, \quad \Lambda_z(\psi_k) = \psi_k \quad \forall |\psi_k\rangle \in \mathcal{B}_z, \\ \Lambda_z(\phi_k) &= \phi_{k \oplus d1}, \quad \Lambda_x(\phi_k) = \phi_k \quad \forall |\phi_k\rangle \in \mathcal{B}_x. \end{aligned} \quad (15)$$

This verifies the assumptions necessary to apply Lemma 4 both to Λ_x, Λ_z , leading to the following:

1. The channels Λ_x, Λ_z are implemented by unitaries U_x, U_z with eigenbases $\mathcal{B}_x, \mathcal{B}_z$
2. The bases $\mathcal{B}_x, \mathcal{B}_z$ are mutually unbiased.
3. The eigenvalues of U_x, U_z are given by the distinct order- d roots of unity ω .

Then, consider the ordering of $\mathcal{B}_z$ through the action of U_x , and likewise $\mathcal{B}_x$ with the ordering from Z . In order to establish simultaneous equivalence of U_x, U_z to (Galois-conjugates of) X, Z , we need to identify the permutations π, σ which associate to each eigenstate the corresponding power of ω . The fact that U_x, U_z act as the clock operators in each other's eigenbases enforces that π, σ are both given as $\pi(j) = \sigma(j) = aj \mod d$ for a fixed $a \in \mathbb{Z}_d^\times$. The ambiguity of a is the manifestation of the Galois symmetries we have previously identified. With a suitably chosen unitary, we find the desired equivalence. Finally, the fact that Λ_f interchanges between $\mathcal{B}_x, \mathcal{B}_z$ can be used to also certify Λ_f as (a Galois-conjugate of) the Fourier-gate via Lemma 13. $\square$

Similar to [17, 19], we have found a way to certify some basic control over the system, which can be further exploited to certify more gates. The first straightforward generalization concerns gates that act by permuting the computational basis.

Corollary 6. *Let $\pi \in \mathfrak{S}_d$ be a permutation, and $P_\pi = \sum_j |\pi(j)\rangle\langle j|$. Then, associating the label ‘p’*

with P_π , the model $\mathcal{F}_d + \{P_\pi\}$ is certified through the inputs $\mathcal{X}_d \cup \{x^j p\}_{j=0}^{d-1} \cup \{fpf\}$ up to $\mathbb{Z}_d^\times$ -symmetries.

Proof. Since we can certify $\mathcal{F}_d$, assume without loss of generality that we have access to state preparations and measurements in the computational and Fourier bases. The sequences $x^j p$ can be used to certify that $\Lambda_p(|j\rangle\langle j|) \stackrel{\cong}{=} |\pi(j)\rangle\langle \pi(j)|$, and likewise fpf for $\Lambda_p(|\omega^0\rangle\langle \omega^0|) = |\omega^0\rangle\langle \omega^0|$. Since $P_\pi|\omega^0\rangle = |\omega^0\rangle$, by Lemma 13 we then have $U_p = P_\pi$. $\square$

Note in this context that, for a permutation P_π as above, we have $P_\pi \in \mathcal{C}\ell(d)$ if and only if $\pi(j) = aj + b \mod d$ for some $a \in \mathbb{Z}_d^\times, b \in \mathbb{Z}_d$ [20]. In particular, any permutation that is not such an affine transformation can be used to reach universality. Before, however, we also need to get the remaining Clifford generator Q into the picture, which is not quite as straightforward, but still doable.

Proposition 7 (Certification of Cl_d). *When d is odd, the input sequences $\mathcal{X}_d \cup \mathcal{X}_q$ with $\mathcal{X}_d$ as in Eq. (14) and*

$$\mathcal{X}_q = \{x^j q, fqxzq^{d-1}f, fqf^2q^{d-1}f\} \quad (16)$$

certify $Cl_d = \mathcal{F}_d + \{Q_d\}$, again up to $\mathbb{Z}_d^\times$ -symmetries.

Proof sketch. For the purpose of this proof, assume that $\mathcal{F}_d$ is correctly implemented, as this can be certified from the inputs $\mathcal{X}_d$ (up to the action of $\text{Gal}(\mathbb{Q}_d)$ of course, but let us assume that $\tau = 1$ in Definition 2 for now). Unitarity of Λ_q can then simply be established by observing $\Lambda_q^d = \mathbb{1}$ on a suitable set of states, using Lemma 13. We therefore limit ourselves to explaining the intuition behind the choice of the input sequences $\mathcal{X}_q$, by demonstrating how the Q -gate, for which we use U_q as alias, is characterized through its interplay with X, Z, F acting on the computational basis. Clearly, $U_q|j\rangle \stackrel{\cong}{=} |j\rangle$ tells us that U_q is diagonal in the computational basis. From $U_q^{d-1}XZU_q|\omega^0\rangle \stackrel{\cong}{=} |\omega^0\rangle$ (cf. Lemma 11) we can infer that $U_q|\omega^0\rangle$ is an XZ -eigenstate, which are of the form $|1; k\rangle = \frac{1}{\sqrt{d}} \sum_j \omega^{-kj+j(j-1)/2} |j\rangle$. In particular, $U_q \stackrel{\cong}{=} \sum_j \omega^{-kj+j(j-1)/2} |j\rangle\langle j| = Z^{-k}Q$ for some k . Now consider the generator relation $Z = Q^{d-1}F^2QF^2$:

$$\begin{aligned} U_q^{d-1}F^2U_qF^2 &= Z^kQ^{d-1}F^2QZ^{-k}F^2 \\ &\stackrel{\cong}{=} Z^k(Q^{d-1}F^2QF^2)Z^k = Z^{2k+1}. \end{aligned} \quad (17)$$

where we used $F^2Z = Z^{-1}F^2$. In particular when d is odd, and we observe $U_q^{d-1}F^2U_qF^2|\omega^0\rangle \stackrel{\cong}{=} |\omega^1\rangle$, we can identify $U_q \stackrel{\cong}{=} Q$. $\square$

5 Robustness considerations

The results Theorem 5, Corollary 6, and Proposition 7 establish that the QSQ-protocol is sound for the considered input sets in the limit of *infinitely many runs*. To ensure the viability of the method as a certification protocol, it is therefore necessary to establish concrete bounds, which connect the model infidelity as defined in Eq. (10) to the average single-round failing probability of this protocol. To be more specific, for a general noisy implementation $\mathcal{N} = (\rho, \{\Lambda_x\}, \{M_a\})$, the average probability of **QSQ** to output ‘accept’ after a single run is given

$$p_{\text{pass}}(\mathcal{N}|\mathcal{X}, \mathbf{a}_{\mathcal{M}}) := \frac{1}{|\mathcal{X}|} \sum_{\mathbf{x} \in \mathcal{X}} \sum_{a \in \mathbf{a}(\mathbf{x})} \text{Tr}[M_a \Lambda_{\mathbf{x}}(\rho)], \quad (18)$$

with $p_{\text{fail}}(\mathcal{N}|\mathcal{X}, \mathbf{a}_{\mathcal{M}}) = 1 - p_{\text{pass}}(\mathcal{N}|\mathcal{X}, \mathbf{a}_{\mathcal{M}})$ as the respective failing probability accordingly.

For the simplest case of single-qubit models, this was done analytically in [17, 18], demonstrating a *linear* dependence. Here, however, similar to the situation in [19], we find that the used methods are largely intractable, and instead repeat the high-level argument derived in [19] which establishes *polynomial* upper and lower bounds as follows:

Theorem 8 (See Thm. 10 in [19]). *Let $\mathcal{M}$ be a d -dimensional quantum model which is certifiable in the sense of Definition 2. Then there exist integer k_1, k_2 and constants C_1, C_2 , such that for any d -dimensional $\mathcal{N}$ isoparametric to $\mathcal{M}$, we can relate the failing probability to $\varepsilon := \min_{\tau \in \text{Gal}(\mathcal{M})} \{\text{inFid}(\tau(\mathcal{M}), \mathcal{N})\}$ as follows:*

$$C_1 \varepsilon^{k_1} \leq p_{\text{fail}}(\mathcal{N}|\mathcal{X}, \mathbf{a}_{\mathcal{M}}) \leq C_2 \varepsilon^{\frac{1}{k_2}}. \quad (19)$$

*In particular, when $\varepsilon > 0$, there exists an integer $N \in \mathcal{O}(\text{poly}(\varepsilon^{-1}) \log(\delta^{-1}))$, such that the corresponding instance of the **QSQ**-protocol outputs ‘reject’ with probability greater than $1 - \delta$.*

Note that due to the non-trivial Galois index, we cannot uniquely identify an implementation. There-

fore, the results above are to be understood with respect to the distance of $\mathcal{N}$ to the closest Galois conjugate of $\mathcal{M}$ up to unitary equivalence.

6 Conclusion

We have demonstrated the versatility of QSQ to cover native gate sets of qudit quantum computing architectures. To do so, we employed a new primitive for self-testing quantum gates based on mutually unbiased eigenbases of operators to establish the necessary coherence properties and identify the spectra. Our results show that while certification via QSQ is not a feature generic quantum models will exhibit, QSQ is well suited to certify gates typically considered in gate-based quantum computing. Additionally, we observed discrete symmetries beyond complex conjugation that must be accounted for, arising as automorphisms of the fields over which the model can be represented. Importantly, we showed that this freedom is independent of the choice of the representation, as long as the underlying fields are reasonable. While the robustness results of [19] also apply in our setting, a proof of a linear dependence along the lines of [17, 18] is necessary to ensure experimental relevance and to link the target fidelity to the sample complexity.

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Appendix

A Qudit Quantum Computing Essentials

In this section, we provide technical details for the statements and claims regarding qudit quantum computation. The first lemma summarizes well-known results about sums of powers of roots of unity, also sometimes known in the literature as Gauss sums. Throughout, all summations are to be understood with respect to the limits 0 and $d - 1$, unless indicated differently.

Lemma 9. Let d be an integer and ω be a d -th order primitive root of unity. Then it holds that

1. $\sum_j \omega^{jk} = 0$ for all $k \in \mathbb{Z}_d^\times$.
2. $\sum_k \omega^{ak^2+bk} = e^{i\theta_a} \sqrt{d} \omega^{-\bar{a}(\bar{2}b)^2}$ for all $a \in \mathbb{Z}_d^\times, b \in \mathbb{Z}_d$ and odd d .

Importantly, in the second identity, the phase $e^{i\theta_a}$ is independent of b and therefore gives important information on how the relative shift of the phase of the expression changes when b is varied.

Both statements in the lemma are or follow from standard results in number theory; see e.g. [33].

The following statements are in themselves relatively straightforward; however, we did not find a good reference citing them in a clean and concise way; therefore, we check them independently.

Lemma 10 (Clifford generator relations). *The generators Q, F of the Clifford and X, Z of the Weyl-Heisenberg group satisfy the following identities: (i) $F^2 = \sum_j |d-j\rangle\langle j|$, (ii) $Q^{d-1}F^2QF^2 = Z$ and (iii) $X = F^3ZF$.*

Proof. We prove the claims one by one with straightforward calculations. We begin with (i):

$$F^2 = \frac{1}{d} \sum_{j,k} \omega^{jk} |j\rangle\langle k| \sum_{l,m} \omega^{lm} |l\rangle\langle m| = \frac{1}{d} \sum_{j,k,l} \omega^{jk+kl} |j\rangle\langle l| = \frac{1}{d} \sum_{j,l} = \frac{1}{d} \sum_{j,l} d \delta_{j+l=0} |j\rangle\langle l| = \sum_j |d-j\rangle\langle j|.$$

Using this relation, we can verify (ii)

$$\begin{aligned} Q^{d-1}F^2QF^2 &= Q^\dagger F^2 Q F^2 = \sum_j \bar{\omega}^{(d-j)(d-j-1)/2} \omega^{j(j-1)/2} |d-j\rangle\langle j| F^2 \\ &= \sum_j \omega^{-j} |d-j\rangle\langle j| F^2 = Z F^2 F^2 = Z. \end{aligned}$$

Finally, we check (iii)

$$F^3ZF = F^\dagger ZF = \frac{1}{\sqrt{d}} \sum_{j,k} \omega^{k+jk} F|k\rangle\langle j| = \frac{1}{d} \sum_{l,m,j,k} \omega^{-lm+k+jk} |l\rangle\langle m|k\rangle\langle j| = \sum_{j,l} \delta_{j+1-l=0} |l\rangle\langle j| = X.$$

□

Lemma 11 (Clifford orbits). *For odd d , the generators of the Weyl-Heisenberg/Clifford group act up to phases in the following way on the Weyl-Heisenberg eigenstates, summarized by the table below.*

	$ j\rangle$	$ \omega^j\rangle$	$ a; j\rangle$		$ j\rangle$	$ \omega^j\rangle$	$ a; j\rangle$
X	$ j+1\rangle$	$ \omega^j\rangle$	$ a; j+a\rangle$	Q	$ j\rangle$	$ 1; j\rangle$	$ a+1; j\rangle$
Z	$ j\rangle$	$ \omega^{j-1}\rangle$	$ a; j-1\rangle$	F	$ \omega^j\rangle$	$ d-j\rangle$	$ d-\bar{a}; \bar{a}(\bar{2}-j)-\bar{2}\rangle$

Here, a refers to the elements of the multiplicative group, $a \in \mathbb{Z}_d^\times$, and $\overline{(\cdot)}$ denotes the multiplicative inverse modulo d . All arithmetic within the kets is to be understood modulo d .

Proof. All calculations aside from the one for $F|a; k\rangle$ are trivial. For the latter we have:

$$\begin{aligned} F|a; k\rangle &= F \frac{1}{\sqrt{d}} \sum_j \omega^{-kj+\bar{2}aj(j-1)} = \frac{1}{d} \sum_{j,l,m} \omega^{ml-kj+\bar{2}aj(j-1)} |m\rangle\langle l|j\rangle \\ &= \frac{1}{d} \sum_{m,j} \omega^{(m-k)j+\bar{2}aj(j-1)} |m\rangle = \frac{1}{d} \sum_m \sum_j \omega^{\bar{2}aj^2+(m-k-\bar{2}a)j} |m\rangle. \end{aligned}$$

From the second statement of Lemma 9 with $a' := \bar{2}a$ and $b' := m - k - \bar{2}a$, we find

$$\begin{aligned} F|a; k\rangle &\stackrel{\cong}{=} \frac{1}{\sqrt{d}} \sum_m \omega^{-2\bar{a}(\bar{2}(m-k-\bar{2}a))^2} |m\rangle = \frac{1}{\sqrt{d}} \sum_m \omega^{-\bar{2}a(m-k-\bar{2}a)^2} |m\rangle \\ &= \frac{1}{\sqrt{d}} \sum_m \omega^{-\bar{2}a(m^2+k^2+(\bar{2}a)^2-2mk-am+ka)} |m\rangle \stackrel{\cong}{=} \frac{1}{\sqrt{d}} \sum_m \omega^{-m(\bar{2}a-\bar{a}k-\bar{2})-\bar{2}am(m-1)} |m\rangle \\ &= \frac{1}{\sqrt{d}} |-\bar{a}; \bar{a}(\bar{2}-k)-\bar{2}\rangle. \end{aligned}$$

□

B Technical results about quantum channels

This section contains several results about quantum channels. We begin with the proof of Lemma 4 as our main technical ingredient enabling 'self-testing', before stating some results already derived in [19] for the convenience of the reader.

Lemma 12 (Variant of Lemma D.6 in [19]). *Let $\Lambda \in \text{CPTP}(\mathcal{H})$ be a channel that maps an ONB $\{|\psi_i\rangle\}_{i \in [d]}$ to another ONB $\{|\phi_i\rangle\}_{i \in [d]}$ of $\mathcal{H}$. Then:*

1. *The channel Λ is unital. In particular it is L^2 -contractive i.e. for any state ρ we have $\text{Tr}[\Lambda(\rho)^2] \leq \text{Tr}[\rho^2]$.*
2. *The states $\{\phi_i\}$ have the unique preimage $\Lambda^{-1}(\phi_i) = \{\psi_i\}$.*
3. *$\Lambda^\dagger(\phi_i) = \psi_i$.*

Proof. Unitality immediately follows, since $\Lambda(\mathbb{1}) = \Lambda(\sum_i \psi_i) = \sum_i \Lambda(\psi_i) = \sum_i \phi_i = \mathbb{1}$. Take a spectral decomposition $\rho = \sum_k q_k \varphi_k$. Now we can apply the Cauchy-Schwarz inequality in the following way:

$$\text{Tr}[\Lambda(\rho)^2] = \sum_{j,k} q_j q_k \text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)] = \sum_{j,k} q_j \sqrt{\text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)]} q_k \sqrt{\text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)]} \quad (20)$$

$$\leq \sqrt{\sum_{j,k} q_j^2 \text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)]} \sqrt{\sum_{j,k} q_k^2 \text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)]} = \sum_{j,k} q_j^2 \text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)]. \quad (21)$$

Since Λ is unital, we know that $\sum_k \Lambda(\varphi_k) = \mathbb{1}$, and thus:

$$\text{Tr}[\Lambda(\rho)^2] \leq \sum_{j,k} q_j^2 \text{Tr}[\Lambda(\varphi_k) \Lambda(\varphi_j)] = \sum_j q_j^2 \text{Tr}[\Lambda(\varphi_j)] = \sum_j q_j^2 = \text{Tr}[\rho^2]. \quad (22)$$

For the second claim, note that since channels cannot create orthogonality, we know that any all preimages of ϕ_i and ϕ_j for $i \neq j$ have to be orthogonal. However, since $\psi_i \in \Lambda^{-1}(\phi_i)$ for all i , we conclude $\Lambda^{-1}(\phi) = \{\psi_i\}$ due to the dimension constraint. Lastly, since $\delta_{ij} = \text{Tr}[\phi_i \phi_j] = \text{Tr}[\phi_i \Lambda(\psi_j)] = \text{Tr}[(\Lambda^\dagger \phi_i) \psi_j]$, we conclude $\Lambda^\dagger(\phi_i) = \psi_i$. □

Lemma 13 (Cor. E.2 and Lemma E.3 in [19]). *Let Λ be a channel and U a unitary defined over the same Hilbert space. Let $\mathcal{B}$ be an ONB of H , and $\mathcal{C}$ a fully coherent set of pure states with respect to that basis. Assume that $\Lambda(\phi) = U\phi U^\dagger$, $\forall |\phi\rangle \in \mathcal{B}$. If moreover $\Lambda(\psi)$ is a pure state for all $\psi \in \mathcal{C}$, then the channel is unitary. Moreover, if the action of Λ and U also coincide on $\mathcal{C}$, then the channel is implemented by U , i.e. $\Lambda(\rho) = U\rho U^\dagger$ for all $\rho \in D(\mathcal{H})$.*

C Proofs of main results

Lemma 14. *Let $\mathcal{M}$ and $\mathcal{N}$ be unitary quantum models on $\mathcal{H} \cong \mathbb{C}^d$. Suppose that there exists $\Theta \in \text{Gl}_{\mathbb{C}}(d)$ such that, componentwise, for all x and a*

$$\Theta U_x \Theta^{-1} = V_x, \quad \Theta M_a \Theta^{-1} = N_a, \quad \Theta |\psi\rangle\langle\psi| \Theta^{-1} = |\phi\rangle\langle\phi|. \quad (23)$$

Then $\mathcal{M}$ and $\mathcal{N}$ are unitarily equivalent. Moreover, if $\Theta \in \text{Gl}_K(d)$ for an algebraic extension $\mathbb{Q} \subseteq K \subseteq \mathbb{C}$, then the implementing unitary can be chosen over a finite algebraic extension of K .

Proof. With $\mathcal{M} = (|\psi\rangle, \{U_x\}_x, \{M_a\}_a)$ and $\mathcal{N} = (|\phi\rangle, \{V_x\}_x, \{N_a\}_a)$, the equivalence relations in Eq. (23) imply,

$$U_x^\dagger \Theta^\dagger \Theta U_x = \Theta^\dagger V_x^\dagger V_x \Theta = \Theta^\dagger \Theta \quad (24)$$

for every x . Similarly, since the measurements are projective and complete,

$$\sum_a M_a \Theta^\dagger \Theta M_a = \sum_a \Theta^\dagger N_a^2 \Theta = \Theta^\dagger \Theta. \quad (25)$$

Because the projectors M_a have rank one, this shows that $\Theta^\dagger \Theta$ is diagonal in the measurement basis. Equivalently, $M_a \Theta^\dagger \Theta = \Theta^\dagger \Theta M_a$ for every a .

An analogous argument applies to the initial state. Let $P = |\psi\rangle\langle\psi|$, $Q = |\phi\rangle\langle\phi|$, $P_\perp = \mathbb{1} - P$, and $Q_\perp = \mathbb{1} - Q$. From $\Theta P = Q \Theta$, it follows that $\Theta P_\perp = Q_\perp \Theta$. Applying the preceding argument to the projective decompositions $P, P_\perp$ and $Q, Q_\perp$ shows that $\Theta^\dagger \Theta$ is block-diagonal with respect to the decomposition induced by P and $P_\perp$. Hence, $\Theta^\dagger \Theta$ commutes with $|\psi\rangle\langle\psi|$. Now define

$$|\Theta| := \sqrt{\Theta^\dagger \Theta}, \quad \Theta_u := \Theta |\Theta|^{-1}. \quad (26)$$

Since Θ is invertible, so is $|\Theta|$. Moreover,

$$\Theta_u^\dagger \Theta_u = |\Theta|^{-1} \Theta^\dagger \Theta |\Theta|^{-1} = \mathbb{1}, \quad (27)$$

and therefore Θ_u is unitary. Since $|\Theta|^{-1} = (\Theta^\dagger \Theta)^{-1/2}$ is a function of $\Theta^\dagger \Theta$, it likewise commutes with all U_x , M_a , and $|\psi\rangle\langle\psi|$. Consequently, for every x ,

$$\Theta_u U_x = \Theta |\Theta|^{-1} U_x = \Theta U_x |\Theta|^{-1} = V_x \Theta |\Theta|^{-1} = V_x \Theta_u. \quad (28)$$

The same argument applies to the measurement operators and the initial state. Therefore,

$$\Theta_u \mathcal{M} \Theta_u^\dagger = \mathcal{N}. \quad (29)$$

Finally, suppose that $\Theta \in \text{Gl}_K(d)$ for an algebraic extension $\mathbb{Q} \subseteq K \subseteq \mathbb{C}$. After adjoining to K the complex conjugates of the entries of Θ , the matrix $\Theta^\dagger \Theta$ is defined over a finite algebraic extension K'/K . Since $\Theta^\dagger \Theta$ is positive definite, it is diagonalizable over a finite algebraic extension of K' . After additionally adjoining the positive square roots of its eigenvalues, $|\Theta|^{-1} = (\Theta^\dagger \Theta)^{-1/2}$, and hence $\Theta_u = \Theta |\Theta|^{-1}$, is defined over a finite algebraic extension of K . $\square$

Proposition 3 (restated). *Let $\mathcal{M} \sim_u \mathcal{N}$ be equivalent unitary quantum models, defined over finite abelian Galois extensions $K_{\mathcal{M}}/\mathbb{Q}, K_{\mathcal{N}}/\mathbb{Q}$, respectively. Then the orbits $\mathcal{O}_{\mathcal{M}}, \mathcal{O}_{\mathcal{N}}$ of $\mathcal{M}$ and $\mathcal{N}$ under the action of $\text{Gal}(K_{\mathcal{M}}/\mathbb{Q}), \text{Gal}(K_{\mathcal{N}}/\mathbb{Q})$ modulo unitary equivalence are bijective.*

Proof. For the purpose of this proof let $K = K_{\mathcal{M}}$, $L = K_{\mathcal{N}}$ and $\mathcal{M} = (|\psi\rangle, \{U_x\}, \{M_a\})$, $\mathcal{N} = (|\phi\rangle, \{V_x\}, \{N_a\})$. Throughout the proof, we repeatedly use the fact that every finite algebraic extension of $\mathbb{Q}$ is contained in a finite Galois extension [29].

The most important part of the proof is showing that we can choose a unitary realizing the equivalence between $\mathcal{M}$ and $\mathcal{N}$ over a suitable algebraic extension of K and L , respectively. Since the fields K and L were defined using the channel superoperators, they need not contain the entries of particular unitary representatives. We therefore first show that we can choose such representatives over finite algebraic extensions. Here, we first consider algebraic extensions of K, L , where we can define the channels through suitable unitaries. Let $U \otimes \bar{U}$ be such a superoperator and choose a nonzero entry u_{ij} of U . The corresponding block is $A_{ij} := \mathbb{1} \otimes \langle i | U \otimes \bar{U} \mathbb{1} \otimes | j \rangle = \bar{u}_{ij} U$. Hence the entries of A_{ij} belong to the field over which $U \otimes \bar{U}$ is defined. Moreover, $A_{ij}^\dagger A_{ij} = |u_{ij}|^2 \mathbb{1}$; therefore, $|u_{ij}|$ is algebraic, and thus we find that U can be defined over an algebraic extension with appropriately adjusted phase.

Since only finitely many algebraic entries are involved, we may choose a finite Galois extension $F/\mathbb{Q}$ containing K, L , and the entries of all $\psi, \phi, M_a, N_a, U_x$, and V_x . By enlarging F if necessary, we may also assume that $\mathbb{Q}(i) \subseteq F$. Since $\mathbb{Q}(i)$ is dense in $\mathbb{C}$, it follows that F is dense in $\mathbb{C}$ with respect to the usual topology.

Let Θ^* be a unitary implementing the equivalence $\mathcal{M} \sim_u \mathcal{N}$. For each instruction x , equivalence of the corresponding unitary channels implies that

$$\Theta^* U_x (\Theta^*)^\dagger = \zeta_x V_x \quad (30)$$

for some phase ζ_x , which is algebraic. After enlarging F if necessary and replacing V_x by $\zeta_x V_x$, which does not alter the corresponding channel, we may assume that $\Theta^* U_x = V_x \Theta^*$, for every x .

We now define the F -vector space of matrices satisfying the componentwise relations between $\mathcal{M}$ and $\mathcal{N}$:

$$\mathcal{V}_F = \{\Theta \in \text{Mat}_{d \times d}(F) \mid \Theta\psi = \phi\Theta; \quad \Theta U_x = V_x \Theta, \forall x; \quad \Theta M_a = N_a \Theta, \forall a\}. \quad (31)$$

Since these equations have coefficients in F , their solution space over $\mathbb{C}$ is the complexification $\mathcal{V}_{\mathbb{C}} := \mathcal{V}_F \otimes_F \mathbb{C}$. The implementing unitary Θ^* is an invertible element of $\mathcal{V}_{\mathbb{C}}$. Let $\{R_1, \dots, R_m\}$ be an F -basis of $\mathcal{V}_F$. Every element of $\mathcal{V}_{\mathbb{C}}$ can then be written as $\sum_{j=1}^m z_j R_j$ with $z_j \in \mathbb{C}$. Since F is dense in $\mathbb{C}$, each z_j can be approximated arbitrarily closely by elements of F . It follows that $\mathcal{V}_F$ is dense in $\mathcal{V}_{\mathbb{C}}$.

Since the determinant is continuous and $\det(\Theta^*) \neq 0$, there exists an open neighborhood O of Θ^* in $\mathcal{V}_{\mathbb{C}}$ consisting entirely of invertible matrices. By density, $O \cap \mathcal{V}_F \neq \emptyset$. Hence, there exists an invertible matrix $\Theta_F \in \mathcal{V}_F$ satisfying $\Theta_F \mathcal{M} \Theta_F^{-1} = \mathcal{N}$ componentwise. By Lemma 14, this equivalence can be implemented by a unitary Θ_F^u defined over a finite algebraic extension E/F . After replacing F by a finite Galois extension of $\mathbb{Q}$ containing E , and retaining the notation F , we may assume that Θ_F^u is defined over F .

Since K, L and F are Galois over $\mathbb{Q}$, the restriction maps $\text{res}_K : \text{Gal}(F) \rightarrow \text{Gal}(K)$ and $\text{res}_L : \text{Gal}(F) \rightarrow \text{Gal}(L)$ are well-defined and surjective (see e.g. Prop. 8.4 of [34], or [35]):

$$\text{Gal}(K) = \text{res}_K(\text{Gal}(F)) \text{ and } \text{Gal}(L) = \text{res}_L(\text{Gal}(F)). \quad (32)$$

Let $\tau \in \text{Gal}(F/\mathbb{Q})$, and denote its restrictions to K and L by τ_K and τ_L , respectively. Since complex conjugation is central in the abelian groups $\text{Gal}(K/\mathbb{Q})$ and $\text{Gal}(L/\mathbb{Q})$, the models $\tau_K(\mathcal{M})$ and $\tau_L(\mathcal{N})$ are again valid unitary models.

Although $\tau(\Theta_F^u)$ need not be unitary, it is invertible. Applying τ to the state and measurement relations gives the corresponding similarity relations between $\tau_K(\mathcal{M})$ and $\tau_L(\mathcal{N})$. For the gates, $\tau(U_x)$ and $\tau(V_x)$ are scalar multiples of unitary representatives of the transformed channels, because the tensor factors of a nonzero channel superoperator are unique up to a scalar. Applying τ to the gate relation and comparing determinants shows that the quotient of these scalars has modulus one. This phase can therefore be absorbed into one of the unitary representatives. Hence, $\tau(\Theta_F^u)$ implements an invertible componentwise similarity between $\tau_K(\mathcal{M})$ and $\tau_L(\mathcal{N})$. By Lemma 14, we therefore also have $\tau_L(\mathcal{N}) \sim_u \tau_K(\mathcal{M})$ for every $\tau \in \text{Gal}(F/\mathbb{Q})$.

Because the restriction maps $\text{res}_K, \text{res}_L$ are surjective (Eq. (32)), we can conclude that there is a bijection

$$\{\tau(\mathcal{M}) \mid \tau \in \text{Gal}(K_{\mathcal{M}})\} / \sim_u \xleftarrow{1:1} \{\tau(\mathcal{N}) \mid \tau \in \text{Gal}(K_{\mathcal{N}})\} / \sim_u, \quad (33)$$

which concludes the proof. $\square$

Theorem 5 (restated). *The input instructions $\mathcal{X}_d := \bigcup_{k=0}^d \{x^k, x^k z, f z^k f, f z^k x f\}$ certify $\mathcal{F}_d$ up to $\mathbb{Z}_d^\times \cong \text{Gal}(\mathbb{Q}_d)$ -symmetries.*

Proof. Let $\mathcal{M} := (\rho, \{\Lambda_x, \Lambda_f, \Lambda_z\}, \{E_k\}_{k \in [d]})$ be a d -dimensional quantum model satisfying $\mathbf{a}_{\mathcal{M}}(\mathbf{x}) \subseteq \mathbf{a}_{\mathcal{F}}(\mathbf{x})$ for all $\mathbf{x}$ in $\mathcal{X}_d$. First of all, since $\mathbf{a}(\mathbf{x}^k) = \{k \bmod d\}$ this implies that the measurement perfectly distinguishes the resulting states $\rho_k := \Lambda_x^k(\rho)$ for $0 \leq k < d$. Due to the dimension constraint, such a behavior is only possible if the ρ_k are orthogonal pure states $\rho_k = \psi_k = |\psi_k\rangle\langle\psi_k|$, and the measurement is the corresponding orthogonal basis measurement, $E_k = \psi_k$. We denote the corresponding orthonormal basis (ONB) as $\mathcal{B}_z = \{|\psi_0\rangle, \dots, |\psi_{d-1}\rangle\}$. Additionally, we have $\rho_0 = \psi_0$ and we further observe that $\psi_d = \Lambda_x^d(\psi_0) = \psi_0$. On the other hand, from the condition $\mathbf{a}(f z^j f) = \{-j \bmod d\}$ we find that the family $\tilde{\rho}_j := \Lambda_z^j \circ \Lambda_f(\psi_0)$ for $0 \leq j < d$ is perfectly distinguishable by the measurement $\{\Lambda_f^\dagger(E_j)\}$. Following the same reasoning as before it follows that the $\tilde{\rho}_j$ are pure orthonormal states $|\phi_j\rangle$, where $\phi_j := \tilde{\rho}_{d-j}$, as well as that $\Lambda_f^\dagger(E_k) = \phi_k$. Again we also find $\phi_0 = \phi_d$. The motivation for the sign reversal in the indexation will become apparent at a later stage of the proof. We refer to this second basis via $\mathcal{B}_x = \{|\phi_0\rangle, \dots, |\phi_{d-1}\rangle\}$.

At this point the relation between the bases $\mathcal{B}_x$ and $\mathcal{B}_z$ is not constrained in any way, they could in principle be equal. However, we observe the following. The conditions $\mathbf{a}(x^k z) = \{-k\}$ and $\mathbf{a}(f z^k x f) = \{k\}$ reveal

that $\Lambda_z(\psi_k) = \psi_k$, and analogously $\Lambda_x(\phi_k) = \phi_k$, respectively. To summarize our findings so far, we have identified two orthonormal bases $\mathcal{B}_x, \mathcal{B}_z$ such that

$$\begin{aligned} \Lambda_x(\psi_k) &= \psi_{k \oplus_d 1} & \text{and} & \quad \Lambda_z(\psi_k) = \psi_k & \text{for all } |\psi_k\rangle \in \mathcal{B}_z, \\ \Lambda_x(\phi_k) &= \phi_k & \text{and} & \quad \Lambda_z(\phi_k) = \phi_{k \ominus_d 1} & \text{for all } |\phi_k\rangle \in \mathcal{B}_x. \end{aligned}$$

This puts us precisely in the situation of Lemma 4 for both channels Λ_x, Λ_z simultaneously, and we conclude the following: First, both channels are implemented by unitaries U_x, U_z , and $\mathcal{B}_x, \mathcal{B}_z$ are their respective eigenbases. In particular, $\mathcal{B}_x, \mathcal{B}_z$ are mutually unbiased. By choosing suitable projective representatives of U_x, U_z , we can achieve that both their eigenvalues are exactly the d different order- d roots of unity and that $|\psi_0\rangle, |\phi_0\rangle$ are the respective eigenvalue-1 eigenstates. That is, for suitable permutations $\pi, \sigma : \mathbb{Z}_d \rightarrow \mathbb{Z}_d$ we can express these unitaries as

$$U_z = \sum_{k \in [d]} \omega^{\pi(k)} \psi_k \quad \text{and} \quad U_x = \sum_{k \in [d]} \omega^{\sigma(k)} \phi_k, \quad (34)$$

where $\pi(0) = \sigma(0) = 0$. Next, we derive constraints on the two permutations π, σ . Since the bases $\mathcal{B}_x, \mathcal{B}_z$ are mutually unbiased, we can suitably modify the phases of the $|\psi_k\rangle$ to express $|\phi_0\rangle = \frac{1}{\sqrt{d}} \sum_{k \in [d]} |\psi_k\rangle$.

Likewise, we adjust the phases of the $|\phi_j\rangle, j \geq 1$ through the canonical relation $|\phi_j\rangle := U_z^{d-j} |\phi_0\rangle$, and obtain the following Fourier-type identities between $\mathcal{B}_x, \mathcal{B}_z$:

$$|\phi_j\rangle = \frac{1}{\sqrt{d}} \sum_{k \in [d]} \omega^{-j\pi(k)} |\psi_k\rangle, \quad |\psi_k\rangle = \frac{1}{\sqrt{d}} \sum_{j \in [d]} \omega^{\pi(k)j} |\phi_j\rangle. \quad (35)$$

Using Eq. (34) and Eq. (35) we can expand the condition $U_x |\psi_b\rangle \stackrel{\circ}{=} |\psi_{b+1}\rangle$ as follows:

$$U_x |\psi_b\rangle = \sum_{j \in [d]} \omega^{\sigma(j)} |\phi_j\rangle \langle \phi_j | \psi_b \rangle = \frac{1}{d} \sum_{j, l \in [d]} \omega^{\sigma(j) + j(\pi(b) - \pi(l))} |\psi_l\rangle \stackrel{\circ}{=} |\psi_{b+1}\rangle. \quad (36)$$

In particular, we conclude that $\left| \sum_{j \in [d]} \omega^{\sigma(j) + j(\pi(b) - \pi(b+1))} \right| = d$, independently of b . Since we chose $\sigma(0) = 0$, this necessitates that

$$\sigma(j) = j(\pi(b+1) - \pi(b)) \quad \text{for all } j, b. \quad (37)$$

Because the LHS of this equation is independent of b , we conclude that $\pi(b+1) - \pi(b) = \text{const} = \pi(1)$ for all b , and therefore $\pi(j) = \sigma(j) = j \cdot \pi(1)$. For the remainder of the work, we abbreviate $\tau := \pi(1)$, representing the corresponding automorphism $\tau \in \text{Gal}(\mathbb{Q}_d)$. Therefore, we find that

$$U_z = \sum_{j \in [d]} \omega^{\tau j} |\psi_j\rangle \langle \psi_j|, \quad U_x = \frac{1}{d} \sum_{j, k, l \in [d]} \omega^{\tau j(k+1-l)} |\psi_l\rangle \langle \psi_k| = \sum_{k \in [d]} |\psi_{k \oplus_d 1}\rangle \langle \psi_k|. \quad (38)$$

Therefore, by considering the gauge unitary $\Theta = \sum_{i \in [d]} |i\rangle \langle \psi_i|$, we find that

$$\Theta U_z \Theta^\dagger = Z^\tau \quad \text{and} \quad \Theta U_x \Theta^\dagger = X, \quad \Theta |\psi_j\rangle \langle \psi_j| \Theta^\dagger = |j\rangle \langle j|. \quad (39)$$

We are left to deal with Λ_f , which is straightforward. Specifically, by testing the inputs $\mathbf{a}_{\mathcal{F}}(\text{fz}^j \mathbf{f}) = \{d-j\}$ we find that $\Lambda_f(\phi_k) = \psi_k$ as well as $\Lambda_f(\psi_0) = \phi_0$ by construction of the $|\phi_j\rangle$. Because $(\mathcal{B}_x, |\psi_0\rangle)$ is fully coherent, from Lemma 13 it follows that Λ_f is unitary and of the form

$$U_f = \sum_{k \in [d]} e^{i\theta_k} |\psi_k\rangle \langle \phi_k|. \quad (40)$$

Because of the Fourier-identities Eq. (35), the condition $U_f |\psi_0\rangle \stackrel{\circ}{=} |\phi_0\rangle$ determines the phases as $e^{i\theta_k} = 1$, and on the other hand, inserting Eq. (35), we obtain:

$$U_f = \sum_{k \in [d]} |\psi_k\rangle \langle \phi_k| = \frac{1}{\sqrt{d}} \sum_{j, k \in [d]} \omega^{\tau k j} |\psi_j\rangle \langle \psi_k|. \quad (41)$$

In particular, we have

$$\Theta U_f \Theta^\dagger = \frac{1}{\sqrt{d}} \sum_{j, l \in [d]} \omega^{\tau j l} |j\rangle \langle l|, \quad (42)$$

and thus $\Theta U_f \Theta^\dagger \otimes \overline{\Theta U_f \Theta^\dagger} = \tau(\mathbf{F} \otimes \overline{\mathbf{F}})$, which concludes the proof. $\square$

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